

AI Engineering — Beginner to Advanced

A 6-Month Project-Driven Curriculum

Goal: Learn to build, deploy, evaluate, and scale production-ready AI applications from scratch.

Core Projects

1. OpenAI API Utilities: summarization, translation, tone transformation, ticket routing, code generation.
2. Customer Support Agent with ticket classification, RAG, response drafting, and escalation.
3. Embeddings Playground for semantic search, recommendations, and classification.
4. YouTube Transcript RAG Chatbot using Pinecone.
5. Enterprise RAG with hybrid search, reranking, guardrails, and RAGAS.
6. Claude Code-like Coding Agent.
7. AI Code Reviewer (CodeRabbit clone).
8. Voice AI Agent (STT → LLM → Tools → TTS).
9. ChatGPT-style Chatbot Platform.
10. Chat with Documents.
11. Perplexity-style AI Search Engine.
12. Lovable/v0-style AI Web App Builder.
13. AI Interviewer.
14. Bonus: AI Trading Research Agent.
15. Deep Learning Mini Projects: TensorFlow Neural Network, MNIST Autoencoder, Multi-input OCR.

Curriculum Modules

0. Python Refresher

Python basics, OOP, error handling, files, environments, type hints, Pydantic, async Python, APIs, NumPy, Pandas.

1. LLM Mental Model

Tokens, context windows, sampling, API anatomy.

2. Conversations

Chat roles, prompts, history, chatbot.

3. Prompt Engineering

Zero/one/few-shot, CoT, templates, structured outputs, context engineering.

4. Real-World LLM Apps

Summarization, translation, support bots, code generation.

5. Production LLM Apps

JSON outputs, retries, batching, token management, FastAPI.

6. Function Calling

Tools API, structured outputs, Pydantic, SDKs.

7. Embeddings

Semantic search, recommendation, classification.

8. Vector Databases

ChromaDB, Pinecone, Milvus.

9. RAG

LangChain, LlamaIndex, hybrid search, reranking, RAGAS.

10. Scalable RAG

Queues, Redis, async workers.

11. Hugging Face

Models, datasets, transformers.

12. Multimodal AI

Vision, audio, VQA, diffusion.

13. AI Agents

Tools, memory, CrewAI, AutoGen, workflows.

14. LangGraph

Stateful workflows, persistence, LangFlow.

15. Memory Systems

Short/long-term memory, Mem0.

16. Graph Memory

Neo4j, Kuzu, Cypher.

17. Voice & Multimodal Agents

Speech and vision agents.

18. MCP & Agent SDK

MCP, Agent SDK, Code Reviewer project.

19. Reliability

SSE, retries, fallback, gateways.

20. Local LLMs

Ollama, Docker, OpenWebUI.

21. Cost Optimization

Prompt caching, KV cache.

22. Observability

LangSmith, Langfuse, tracing.

23. Evaluation

LLM-as-judge, RAGAS.

24. Security

OWASP LLM Top 10, guardrails.

25. MLOps & LLMOps

Deployment, monitoring, governance.

26. Production Best Practices

Moderation, testing, AI Trading Agent.

27. Math & PyTorch

Linear algebra, tensors, optimization.

28. Computer Vision

CNNs, augmentation.

29. Sequence Models

RNN, LSTM, GRU, OCR.

30. Transformers

Attention, GPT, BERT, tokenizer.

31. Generative Models

GANs, VAEs.

32. LLM Internals

Flash Attention, MoE, RLHF, quantization.

33. Fine-Tuning

LoRA, QLoRA, Unsloth, ViTs.

Audience

Software engineers, backend developers, AI enthusiasts, students, and professionals who want to build production AI systems.

Expected Outcomes

- Build production-ready AI applications.
- Design RAG systems.
- Build AI agents with tools and memory.
- Deploy local and cloud-hosted LLMs.
- Add observability, evaluation, and guardrails.
- Understand deep learning and LLM internals.
- Fine-tune models using LoRA/QLoRA.
- Build a strong AI engineering portfolio.

Overall Assessment

This is an excellent practical curriculum that emphasizes production AI engineering, modern RAG systems, agentic workflows, observability, evaluation, and deployment. It pairs best with a solid software engineering foundation.